

PHYSICS OF COMPLEX SYSTEMS

LECTURE AND TUTORIALS – PROF. DR. HAYE HINRICHSSEN – MAXIMILIAN SCHÄFFERER – SS 2026

```
In[1]:= Attributes[CircleTimes] = {Flat, OneIdentity};
CircleTimes[a_List /; VectorQ[a], b_List /; VectorQ[b]] :=
  Flatten[KroneckerProduct[a, b]];
CircleTimes[a_List /; MatrixQ[a], b_List /; MatrixQ[b]] :=
  KroneckerProduct[a, b];

In[4]:= {a, b} ⊗ {c, d, e}
Out[4]:= {a c, a d, a e, b c, b d, b e}

In[5]:= SX = {{0, 1}, {1, 0}};
SX ⊗ SX
Out[6]:= {{0, 0, 0, 1}, {0, 0, 1, 0}, {0, 1, 0, 0}, {1, 0, 0, 0}}
```

Automated tensor product in Mathematica

EXERCISE 5.1: TENSOR PRODUCT

(6P)

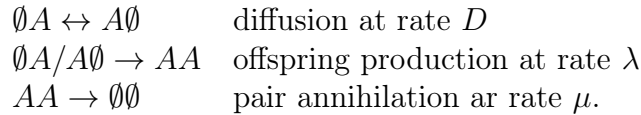
Let $\mathbf{A}$ and $\mathbf{B}$ be square matrices acting on the vector space V .

- Prove that $e^{\mathbf{A} \otimes \mathbb{1} + \mathbb{1} \otimes \mathbf{B}} = e^{\mathbf{A}} \otimes e^{\mathbf{B}}$. (2P)
- Consider the eigenvectors and eigenvalues of $\mathbf{A}|a\rangle = a|a\rangle$ and $\mathbf{B}|b\rangle = b|b\rangle$. What are the eigenvalues and eigenvectors of $\mathbf{A} \otimes \mathbf{B}$? (1P)
- Compute the commutators of the Pauli matrices $[\sigma_i, \sigma_j]$ and $[\sigma_i \otimes \sigma_i, \sigma_j \otimes \sigma_j]$. (1P)
- Compute $e^{\phi \sigma_i \otimes \sigma_j}$ for $\phi \in \mathbb{R}$. (2P)

EXERCISE 5.2: CREATION-ANNIHILATION MODEL

(6P)

Consider a one-dimensional chain of three sites which are either empty ($\emptyset$) or occupied by a single particle (A), assuming closed (non-periodic) boundary conditions. On neighboring sites the particles diffuse and react with each other according to the following rules:



- Construct the 4×4 interaction matrix of the two-site Liouville operator $\mathcal{L}^{(2)}$ in the canonical configuration basis $(\emptyset \emptyset, \emptyset A, A \emptyset, AA)$. (2P)
- Determine the 8×8 matrix of the Liouville-Operator

$$\mathcal{L}^{(3)} = \mathbb{1} \otimes \mathcal{L}^{(2)} + \mathcal{L}^{(2)} \otimes \mathbb{1}$$

acting on a chain with three sites ($\mathbb{1}$ denotes the 2×2 unit matrix). (2P)

- Consider the special case $\lambda = 0$ and calculate the eigenvalues of $\mathcal{L}^{(3)}$. Why is the ground state two-fold degenerate? (2P)

($\Sigma = 12P$)

Please submit your solution electronically as a single PDF file until Wednesday, May 20, 2026, via Wue-Campus.